YuJia Liang

+886 965 160 695 | ch993115@gmail.com | LinkedIn | GitHub | yujialiang.com | Taipei, Taiwan

Education

National Taiwan Ocean University Keelung, Taiwan

B.S. in Computer Science and Engineering *Sep. 2022 – Jun. 2026*

Relevant Courses: Data Structures & Algorithms, Graph Theory Algorithms, Operating Systems, Database Systems, Computer Networks, Machine Learning, Natural Language Processing, Generative AI, Object-Oriented Programming, Cyber Security, Linear Algebra, Discrete Mathematics, Probability Theory

Experience

ITRI (Industrial Technology Research Institute) Jun. 2025 – Aug. 2025

AI Image Recognition Intern, Electronics and Optoelectronics Research Laboratories *Hsinchu, Taiwan*

- Optimized multi-task YOLO model for autonomous driving, improving inference speed while maintaining driving area accuracy and detection recall
- Implemented real-time image streaming and detection result transmission via Ethernet/ROS, processing 30 FPS with sub-50ms latency
- Applied neural network pruning techniques to YOLO models for edge deployment
- Supported research initiatives by reproducing and exploring state-of-the-art (SOTA) models to benchmark performance and identify potential improvements

Research & Projects

RiffNode – AI Guitar Effects Studio 2026

SwiftUI, AVFoundation, Accelerate/vDSP *WWDC26 Swift Student Challenge*

- **Selected as an Apple WWDC26 Swift Student Challenge Winner** – one of only 8 winners in Taiwan and 350 worldwide – for a professional guitar effects app built entirely with native Apple frameworks (no third-party APIs), compressing to under 2 MB; featured in national press (UDN, CNA)
- Built a dual-AI tone engine: Apple Foundation Models generates structured DSP parameters on-device via `@Generable` while Gemini 2.0 Flash enriches explanations, staying fully functional offline
- Engineered a low-latency real-time audio pipeline with `AVAudioEngine` driving 11 professional effects (compressor, 10-band EQ, overdrive, distortion, reverb, delay, modulation) in a drag-and-drop signal chain with live reconnect
- Implemented hands-free control using Vision Framework face-landmark detection (nod, tilt, mouth-open gestures) plus real-time Accelerate/vDSP FFT analysis, autocorrelation pitch detection, tuner, metronome, and chord-aware tone suggestions

Volleyball Performance Analysis System – Senior Capstone Nov. 2024 – Dec. 2025

Python, PyTorch, YOLO, OpenCV, React, TypeScript, FastAPI, Celery, Redis, Docker, pytest Advisor: Prof. Pei-Yi Ding

- Developed comprehensive volleyball analytics system achieving 94.49% mAP@0.5 for action recognition (5 classes: serve, spike, block, receive, set) and 79.5% ball tracking accuracy
- Architected distributed microservices with FastAPI backend, Celery task queue, and Redis message broker; implemented WebSocket endpoints for real-time analysis progress tracking
- Built multi-model deep learning pipeline: VballNet (U-Net ONNX) for ball tracking at 200+ FPS, YOLOv11m for action detection, YOLOv8+Norfair for player tracking with 87.6% ID consistency
- Developed React/TypeScript frontend with Video.js player, Recharts analytics dashboard, interactive timeline with frame-level seeking, and real-time bounding box overlays
- Implemented trajectory filtering pipeline with velocity-based outlier removal, Gaussian smoothing, and polynomial interpolation; action consolidation algorithm reducing fragmented events by 45%
- Containerized application with Docker Compose orchestrating 5 services; configured GitHub Actions CI/CD with flake8 linting and automated builds
- Wrote comprehensive test suite using pytest with async support, achieving coverage across API endpoints, database operations, and ML inference pipelines

Ghote Notes – AI Note-Taking Application Nov. 2025 – Present

Tauri 2.0, Rust, React 19, TypeScript, Lexical, D3.js, Supabase, PartyKit, Yjs, Vitest, Playwright *Individual Project*

- Built privacy-focused macOS note-taking app using Tauri 2.0 (Rust) with React 19 frontend, featuring vault-based storage with Obsidian-compatible markdown files
- Implemented rich text editing with Facebook's Lexical framework, supporting slash commands, KaTeX math equations, Mermaid diagrams, and bidirectional markdown sync

- Developed interactive knowledge graph using D3.js with wiki-style [[link]] parsing, enabling visual exploration of note relationships
- Built local RAG system using Transformers.js with all-MiniLM-L12-v2 embeddings (384-dim) for semantic search, featuring smart chunking and cosine similarity ranking
- Implemented real-time collaboration via PartyKit WebSocket server with Yjs CRDT for conflict-free concurrent editing and cursor presence
- Integrated Supabase cloud sync with PostgreSQL RPC functions and multi-device vault synchronization
- Achieved 88.75% test coverage with 521 unit tests using Vitest; built comprehensive Tauri API mocks for file system, dialogs, and path utilities

SyncUp – Social Calendar App 2025

Flutter, Dart, Firebase, Google Maps API, Google Gemini API *Individual Project*

- Developed social calendar app with real-time sharing and friend management using Flutter and Firebase
- Implemented location-based event suggestions with Google Maps integration and Firebase Authentication
- Built AI-powered chatbot connected with Google Gemini to intelligently generate schedules and suggest meetup opportunities based on user preferences

Technical Skills

Programming Languages: Python, Java, C++, JavaScript/TypeScript, Rust, Swift, Dart, SQL
ML/AI: PyTorch, TensorFlow, OpenCV, ONNX Runtime, scikit-learn, NumPy, Pandas, CreateML, Foundation Models, Vision Framework, Transformers.js, Norfair
Frameworks: React, Flutter, FastAPI, Node.js, SwiftUI, AVFoundation, Accelerate/vDSP, Tauri, Lexical, Celery, Video.js, Recharts, D3.js
DevOps & Tools: Git, Docker, GitHub Actions, pytest, Playwright, Vitest, Redis, PostgreSQL, SQLite, Supabase, Firebase
Languages: Mandarin (Native), English (Fluent, TOEFL 110, TOEIC 920), Japanese (JLPT N3)

Leadership & Activities

Volleyball Department Team Captain 2022 – present

Keelung, Taiwan

NTU Photography Club (NTUPhoto) 2025 – 2026

Member & Darkroom Officer *Taipei, Taiwan*

NTUFC Kronos Quantitative Reading Club Sep. – Dec. 2025

Hackathon Participant 2022 – present